Anomaly Detection of Hyperspectral Image With Hierarchical Antinoise Mutual-Incoherence-Induced Low-Rank Representation

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***Abstract*— Hyperspectral image (HSI) anomaly detection (AD) generally considers background pixels as low-rank distribution and anomaly pixels as sparse distribution. However, it is usually difficult to construct an accurate background dictionary for the background pixels composed of different land covers, and completely separate sparse anomaly targets from various complicated background pixels with complex mixed noise interference. To address these challenges, we propose an antinoise hierarchical mutual-incoherence-induced discriminative learning (AHMID) method for the AD of HSI. A structural incoherence constraint is designed to constrain the inherent dissimilarity and incoherence between the background and anomalies for improving their separability. Then, a first-order statistic constraint is conducted on targets to enhance the anomaly representation, and a decentralization constraint is used on the background to suppress the background representation. Meanwhile, a mixed noise model is constructed by $\ell_{1,1}$-norm and Frobenius norm to improve the antinoise performance. Finally, a hierarchical alternative strategy is developed to gradually optimize the background and anomalies. Experiments on six HSI AD datasets show that the proposed method outperforms a few state-of-the-art AD algorithms. Code: https://github.com/HalongL/HAD-AHMID**

***Index Terms*— Antinoise, hierarchical structure, hyperspectral anomaly detection (AD), incoherent constraint, low-rank representation (LRR).**

Manuscript received 25 January 2023; revised 18 March 2023; accepted 8 April 2023. Date of publication 21 April 2023; date of current version 10 May 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62201109, Grant 62071340, and Grant 62276037; in part by the Natural Science Foundation of Chongqing under Grant CSTB2022NSCQ-MSX0452; in part by the Key Scientific and Technological Innovation Project for "Chengdu-Chongqing Double City Economic Circle" under Grant KJCXZD2020025; and in part by the Macao Young Scholars Program under Grant AM2020008. *(Corresponding author: Fulin Luo.)*

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Digital Object Identifier 10.1109/TGRS.2023.3269097

I. Introduction

HYPERSPECTRAL remote sensing is currently a useful and popular earth observation technique for identifying land covers [1], [2]. Compared with traditional panchromatic and multispectral remote sensing images, hyperspectral images (HSIs) can provide almost continuous spectral information to reveal the characteristics of the ground object [3], [4], [5]. In HSI, the spectral resolution can reach 10 nm, and it possesses dozens, tens, or even hundreds of spectral bands, which generates a big difference between different land covers and has unique advantages for target detection and classification [6], [7]. Target detection is considered as a binary classification problem aiming to separate specific target pixels from background pixels including multiple land covers [8], [9]. In target detection, a significant application is anomaly detection (AD). It does not use any prior information depending only on the feature difference of HSI to discriminate the anomaly targets from background pixels composed of several different land covers, which has been widely used in both civil and military applications such as rare mineral exploration and camouflage detection [10], [11].

There are many methods for hyperspectral AD, which are divided into two main categories: statistics-based methods and geometric modeling-based methods. One classic approach uses a statistical property of HSI to detect anomaly information and assumes that background pixels obey a certain statistical criterion that anomaly pixels do not match. The Reed–Xiaoli (RX) detector is a well-known AD method, which uses a Gaussian model to estimate anomaly pixels based on the global mean and covariance of HSI [12]. RX uses global statistical information, which is not suitable for complex background distribution. Thus, a local RX (LRX) was designed to adaptively detect anomalies under multiple land-cover backgrounds. Unlike RX, LRX estimates the background through a local window around a test pixel. To further enhance the separability between the backgrounds and anomalies, a kernel RX was proposed by transforming the original data into a kernel space where data have better discrimination performance. In addition, a subspace RX (SSRX) was suggested to reduce the dimensionality of HSI while enhancing the discriminating performance between the backgrounds and anomalies [13]. Following further development, some advanced

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statistics-based methods have been designed for AD of HSI such as support vector data description (SVDD) [14], kernel isolation forest detector (KIFD) [15], and two-step generalized likelihood ratio test (2S-GLRT) [16].

The above statistics-based methods depend on a certain statistical rule to detect the anomalies of HSI, whereas it is very difficult to find a proper model to describe the statistical distribution for complex HSI. Geometry-based modeling approaches are proposed to reflect the intrinsic properties of HSI, which can be further divided into two subclasses: representation-based and matrix decomposition-based. Collaborative representation assumes that the background can be collaboratively represented by its neighbors, which is not suitable for anomalies [17], [18]. Based on this assumption, a collaborative representation detection (CRD) method was proposed based on the concept that each background pixel can be represented by a linear combination of its neighbor pixels while inapplicable to anomalous pixels [19]. Sparse representation (SR)-based algorithms [20], [21] are very popular for HSI target detection, which stems from linear mixture models of HSI. It assumes that each pixel in HSI can be sparsely represented by an overcomplete dictionary containing both target and background spectra. SR models have also been applied to AD. Ling et al. [22] proposed an SR method based on a linear mixed model that attaches explicit physical meaning to the solution. Although the above methods can describe the relationship between different pixels, none of these methods considers the correlation of all pixels in the HSI. Therefore, the global information is not considered in these methods, leading to a weak description of the overall structure of the HSI.

To make full use of the structural information of the whole HSI, matrix decomposition methods have been favored by many researchers. Recently, low-rank sparse matrix decomposition (LSDM) has been successfully applied to hyperspectral AD [23], [24]. It considers the backgrounds to be low-rank distributions and the anomalies to be sparse distributions [25], [26]. Robust principal component analysis (RPCA) [27] is one of the most popular LSDM-based methods for AD, which can decompose the HSI data into the low-rank and sparse components that, respectively, correspond to the backgrounds and anomalies. Qu et al. [25] proposed an AD method based on low-rank decomposition, which constructs a dictionary based on a lot of remarkable features. Zhang et al. [28] applied the low-rank prior knowledge of the background to compute the background statistics and constructed a Mahalanobis-distance-based anomaly detector, called LSMAD. In [24], a low-rank and sparse decomposition with mixed Gaussian model (LSDM-MoG) was applied for HSI AD, which considers the sparse component as a mixture of Gaussian. Considering that the background in HSI consists of different kinds of materials, low-rank representation (LRR) [29] is subsequently introduced to model the background from multiple subspaces. This approach portrays the relationship of all HSI pixels from a global perspective while neglecting local information. To address this issue, Xu et al. [30] constructed a low-rank and SR (LRASR) anomaly detector, which designed a clustering-based dictionary construction algorithm to explore global and local structural information. Tan et al. [31] added a spatial constraint and adopted multiple local windows (MLW_LRRSTO) to enhance the utilization of spatial information. In addition, to consider the smooth prior in the backgrounds and the structural features of HSI, some advanced methods have been developed for AD, such as graph and total variance regularized LRR (GTVLRR) [32], local spatial constraint and total variation (LSCTV) [33], and enhanced total variance with endmember background dictionary (ETVEBD) [34].

HSIs have the characteristics of difficult background and target separation, noise interference, and depth hiding of some features for AD. By reviewing the aforementioned AD methods in HSI, there are several key issues that remain unresolved. First, the discriminative ability of background dictionary and anomaly is not well revealed and utilized for AD, and the background is often mixed with the anomaly target. Second, a number of different types of noise are inevitably presented in HSI due to the complexity of the environment, such as atmospheric interference and sensor noise. Finally, the inadequacy of single learning limits the discriminative performance of features, and the sparse components of the model decomposition inevitably contain noise.

To address the above issues, we propose to accurately model the three key basic components, i.e., background, anomalies, and noise, through a newly developed hierarchical mutual-incoherence-induced discriminative learning strategy. The structural incoherence constraint, the first-order statistic constraint, and decentralization constraints are used to separate the background from the anomaly, then the noise is modeled as a mixture, and finally a hierarchical structure is used to address the problem of inadequate single learning. The proposed method can enhance the discrimination of background and anomaly with the removal of complex noise. The main highlights and contributions of this work can be briefly described as follows.

1) To enhance the separability of the background and the anomalies, an incoherence-induced discriminative learning strategy is proposed to boost the structural incoherence between the background and anomaly components, with the first-order statistics on the anomalies to enhance the anomaly target, and the decentralization constraint on the background to suppress the background.
2) In real HSI, it may contain complex mixed noise from the imaging process. A joint regularization containing $\ell_{1,1}$-norm and Frobenius norm is adopted to accurately model and portray the complex noise. Since the background and the noise are often mixed in the anomalies, we further use the first-order statistics to eliminate them for achieving strong antinoise performance.
3) A hierarchical structure is designed to gradually separate the low-rank backgrounds and the sparse anomalies, which can further strengthen the anomalies and suppress the background. The proposed method makes full use of the characteristics of the background and anomalies in HSI for improving the discriminative performance to achieve a high detection rate and a low false alarm rate.

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The rest of this article is organized as follows: Section II introduces the LRR for AD. Section III describes the proposed method in detail. In Section IV, experiments are conducted and analyzed. Section V concludes this article.

II. Low-Rank Representation

The observed HSI dataset $\mathbf{X} \in \Re^{D\times N}$ has N pixels with D spectral bands. The preconstructed background dictionary is denoted by $\mathbf{D} \in \Re^{D\times N_b}$ with N_b atoms. In HSI, a key prior knowledge for anomalous pixels is that they should be different from their surrounding background pixels. In addition, due to the universality of the background pixels, there is usually a strong correlation between different background pixels, i.e., background pixels can be represented by some other surrounding similar background pixels. Thus, the HSI data matrix can be preliminarily decomposed into two components: background and anomalies. The basic decomposition model can be expressed as

$$\mathbf{X} = \mathbf{DZ} + \mathbf{S} \tag{1}$$

where $\mathbf{DZ}$ denotes the background component in HSI, $\mathbf{D} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_{N_b}]$ is the background dictionary consisting of representative background pixels (N_b is the total number of atoms in the dictionary), $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_N]$ denotes the representation coefficients, and $\mathbf{S} = [\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_N]$ denotes the anomaly component.

Obviously, problem (1) has many feasible solutions. To obtain a meaningful solution, some more prior regularizations are needed to, respectively, characterize the background and anomalies. The background component has abundant similar and correlated samples, requiring the corresponding coefficient matrix to be low-rank. Thus, the decomposition is defined as

$$\min_{\mathbf{Z},\mathbf{S}} \ \text{rank}(\mathbf{Z}) + \alpha\|\mathbf{S}\|_{2,0} \quad \text{s.t. } \mathbf{X} = \mathbf{DZ} + \mathbf{S} \tag{2}$$

where $\text{rank}(\mathbf{Z})$ is the rank of representation matrix $\mathbf{Z}$, $\|\cdot\|_{2,0}$ is used to measure the sparsity of $\mathbf{S}$, and $\alpha > 0$ is the tradeoff parameter controlling the contribution of $\mathbf{S}$.

However, (2) is a nonconvex optimization problem. Thus, the low-rank approximation of a matrix can be solved by minimizing its nuclear norm under certain conditions. In addition, the sparse anomalous component can be constrained by the $\ell_{2,1}$-norm. Therefore, (2) can be transformed as

$$\min_{\mathbf{Z},\mathbf{S}} \ \|\mathbf{Z}\|_* + \alpha\|\mathbf{S}\|_{2,1} \quad \text{s.t. } \mathbf{X} = \mathbf{DZ} + \mathbf{S} \tag{3}$$

where $\|\cdot\|_*$ denotes the nuclear norm of matrix and $\|\mathbf{S}\|_{2,1} = \sum_{j=1}^{n}(\sum_{i=1}^{n}([\mathbf{S}]_{ij})^2)^{1/2}$ is called the $\ell_{2,1}$-norm, which could encourage the columns of $\mathbf{S}$ to be zero. $\alpha > 0$ is used to balance the effects of the two parts, which could be chosen according to the properties of the two norms, or tuned empirically. The anomalous response $A(x_i)$ of a pixel can be obtained by calculating the ℓ_2-norm of each column in $\mathbf{S}$, and it can be represented as

$$A(x_i) = \left\|[\mathbf{S}]_{:,i}\right\|_2 = \sqrt{\sum_j \left([\mathbf{S}]_{j,i}\right)^2} \tag{4}$$

where $\|[\mathbf{S}]_{:,i}\|_2$ is the ℓ_2-norm of the ith column of $\mathbf{S}$. If it is greater than a threshold value, $\mathbf{x}_i$ is an anomalous pixel; otherwise, it is a background pixel.

III. Proposed Method

A. Overall Model

HSIs consist of background, anomalies, and noise. This article proposes an overall architecture to accurately portray the image data and provides a feasible way to separate the background and the anomalous targets as much as possible

$$\min_{\mathbf{Z},\mathbf{D},\mathbf{S},\mathbf{E}} \ \|\mathbf{Z}\|_* + \|\mathbf{S}\|_{2,1} + \alpha\Psi(\mathbf{D},\mathbf{S}) + \lambda\Theta(\mathbf{E}) \quad \text{s.t. } \mathbf{X} = \mathbf{DZ} + \mathbf{S} + \mathbf{E} \tag{5}$$

where it includes the low-rank term $\mathbf{Z}$, the sparse anomaly target term $\mathbf{S}$, the unified constraint terms $\Psi(\cdot)$, and the noise term $\Theta(\cdot)$. α and λ are two tradeoff parameters.

According to (5), different constraint terms can be used to separate the background and the anomalies, and model the noise. For anomaly $\mathbf{S}$, once $\Psi(\cdot)$, $\Theta(\cdot)$, and the background dictionary $\mathbf{D}$ are given, $\mathbf{S}$ can be solved. For background dictionary $\mathbf{D}$, a new background dictionary can be solved and updated by the above solved $\mathbf{S}$. In this way, a hierarchical structure is achieved to further update the low-rank background and the sparse anomalies. Specifically, the whole process can be described by Fig. 1, we first use a preconstructed background dictionary to obtain the anomaly component based on the S-model. Then, we construct an alternative updating layer, which updates the background dictionary based on the D-model and reoptimizes the anomaly component based on the S-model. With several alternative updating layers, we can achieve the proposed antinoise hierarchical mutual-incoherence-induced discriminative learning (AHMID) method.

B. Target Enhancement Model

In general, HSIs not only have background and anomalies, but also contain complex mixed noise. The three basic components are mixed to form an HSI. The mixing between the background and the anomalies in HSI is very difficult to be separated. Here, we propose a discriminative learning strategy for the background and the anomalies. Inspired by [35], [36], and [37], we consider the incoherence between the low-rank background and the sparse anomalies. Moreover, the sparse anomalies may contain a few backgrounds and noise, which are closer to the first-order statistic distribution. The incoherence expects to produce a low-rank matrix and a sparse matrix as independent as possible. Thus, the common features between the low-rank background and the sparse anomalies are suppressed, whereas the independent features are preserved. Therefore, our approach can provide additional discriminative power to the original LRR model by facilitating the structural incoherence of the two key components, and the detection performance is expected to be improved.

Based on (5), we define $\Psi(\cdot)$ with the joint constraint of structural incoherence and first-order statistic on the sparse anomalies, which enhances the inconsistency between the

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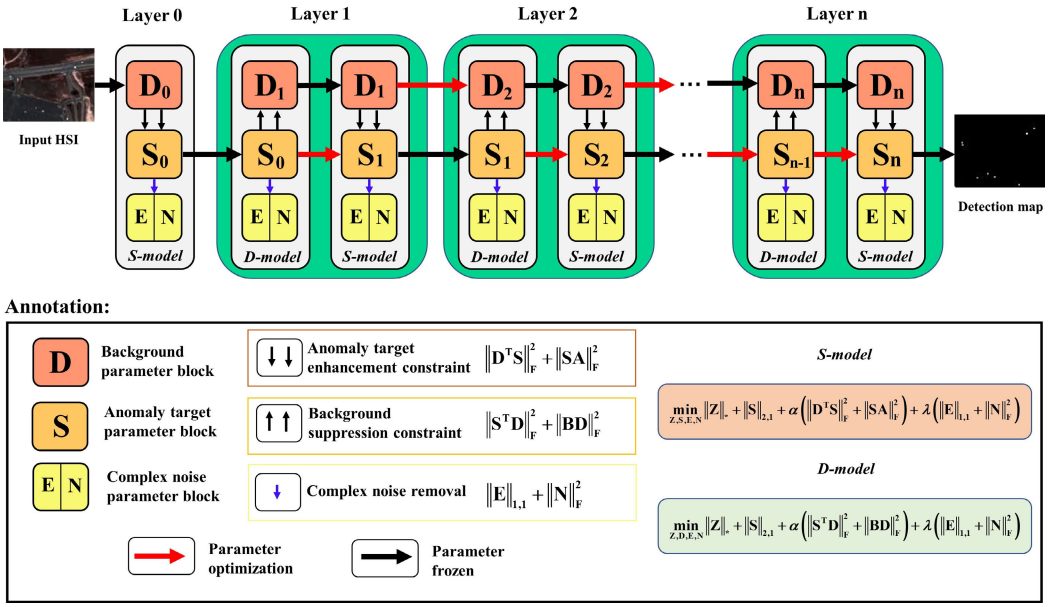


Fig. 1. Illustration of the proposed AHMID method.

low-rank background dictionary and the sparse anomaly matrix, and eliminates a few backgrounds and noise in the anomalies. $\Psi(\cdot)$ is defined as

$$\Psi(\mathbf{D},\mathbf{S}) = \left\|\mathbf{D}^{\mathbf{T}}\mathbf{S}\right\|_F^2 + \|\mathbf{SA}\|_F^2 \tag{6}$$

where $\|\mathbf{D}^T\mathbf{S}\|_F^2$ is the Frobenius norm between the background dictionary and the anomaly components to encourage the structural incoherence between $\mathbf{D}$ and $\mathbf{S}$. $\|\mathbf{SA}\|_F^2$ is the first-order statistical representation of $\mathbf{S}$ to enhance the representation performance of anomaly targets, and $\mathbf{A} \in \Re^{N\times N}$ is a matrix with all elements of $1/N_b$.

C. Background Suppression Model

In AD, background suppression is also very important, that is, the less targets are mixed in the background, the better the background suppression is. Thus, we consider the universal background dictionary to suppress the background and design the following decentralization of the background dictionary $\mathbf{D} = (\mathbf{d}_1, \ldots, \mathbf{d}_{N_b})^T$, because the decentralization operation can remove the influence of the outlier background and enhance the universality of the background dictionary. According to $\overline{\mathbf{D}^{\mathbf{T}}} = (1/N_b)\sum_{i=1}^{N_b} d_i$, we can obtain

$$\begin{aligned}(\mathbf{D}^{\mathbf{T}} - \overline{\mathbf{D}^{\mathbf{T}}})^T &= \left([d_1, \ldots, d_{N_b}]\left[\mathbf{I}_D - \frac{1}{N_b}\mathbf{E}_D\right]\right)^T \\ &= \left(\mathbf{D}^{\mathbf{T}}\left[\mathbf{I}_D - \frac{1}{D}\mathbf{E}_D\right]\right)^T = \mathbf{BD}\end{aligned} \tag{7}$$

where $\mathbf{B} = (\mathbf{I}_D - (1/D)\mathbf{E}_D)^T$, $\mathbf{D}$ is the background dictionary, $\mathbf{I}_D$ is the identity matrix, and $\mathbf{E}_D$ is a square matrix with all elements of 1.

Similarly replacing $\Psi(\cdot)$ in (5) with a joint constraint with the structural inconsistency and the centralization of the background dictionary $\mathbf{D}$, this not only enhances the discrimination between the low-rank background dictionary and the sparse anomalies, but also suppresses the outlier background. $\Psi(\cdot)$ is denoted by

$$\Psi(\mathbf{D},\mathbf{S}) = \left\|\mathbf{S}^{\mathbf{T}}\mathbf{D}\right\|_F^2 + \|\mathbf{BD}\|_F^2 \tag{8}$$

where $\|\mathbf{S}^T\mathbf{D}\|_F^2$ is the Frobenius norm between the background dictionary and the anomaly components to encourage the structural incoherence between $\mathbf{S}$ and $\mathbf{D}$. $\|\mathbf{BD}\|_F^2$ is a decentralized representation of the background dictionary $\mathbf{D}$ to suppress the outlier background.

D. Mixed Noise Model

Due to the imaging process, such as atmospheric disturbance and sensor noise, different types of noise might be inevitably presented in HSI. As a result, a single noise distribution cannot accurately model the complex noise, which limits the performance of noise suppression. In practical applications, HSI generally contains some Gaussian noise to impact the performance of AD; meanwhile, the observed images are captured by means of scanning imaging that may achieve a lot of strip noise. Therefore, noise in HSI is composed of various types mainly including Gaussian noise and strip noise. In AD, we must construct the mixed noise model to remove the influence on the discriminative performance of anomalies. Thus, we should consider the mixed noise including Gaussian noise and strip noise in our model. Inspired by [38], we use the $\ell_{1,1}$ and Frobenius norm to characterize the strip noise and the Gaussian noise, and the noise term can be reformulated as follows:

$$\Theta(\mathbf{E}) = \left(\|\mathbf{E}\|_{1,1} + \|\mathbf{N}\|_F^2\right). \tag{9}$$

According to (9), the complex mixed noise model can promote the anomalies separation from the background and decrease the influence of noise. The AD result can be obtained by the model (6) with a preconstructed background dictionary $\mathbf{D}$ in one step. Thus, model (6) is a one-step detection

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model. However, due to the deep mixing of spectra, such a shallow leaning structure might not be able to completely separate the HSI into background, anomalies, and noise. In addition, the fixed preconstructed background dictionary $\mathbf{D}$ has a great impact on the detection effect, that is, if the constructed background dictionary is not ideal, the AD ability will be reduced.

E. Hierarchical Alternative Updating

For a one-step detection strategy, it is very difficult to reveal the intrinsic features of HSI to improve the detection performance, and the quality of the background dictionary is critical for detection accuracy. If the background dictionary is contaminated with anomalies or noise, it is difficult to produce the desired performance for AD. To address this issue, a hierarchical structure is designed to gradually separate the background and the anomalies. This model does not rely on the quality of the initialized background dictionary and can gradually reveal the intrinsic discriminative background and anomaly components of HSI. Therefore, we fix the anomaly targets $\mathbf{S}$ from model (6) and update the background dictionary $\mathbf{D}$ based on model (8) to enhance the suppression of the background. Then, we fix the background dictionary $\mathbf{D}$ to update the anomaly matrix $\mathbf{S}$ based on model (6).

Therefore, based on models (5), (6), (8), and (9), we construct a hierarchical structure learning strategy, which has two models including the S-model to obtain the anomalies, and the D-model to update the background dictionary. They are represented as

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{S},\mathbf{E},\mathbf{N}}\ & \|\mathbf{Z}\|_* + \|\mathbf{S}\|_{2,1} + \alpha\left(\left\|\mathbf{D}^{\mathbf{T}}\mathbf{S}\right\|_F^2 + \|\mathbf{SA}\|_F^2\right) \\ & + \lambda\left(\|\mathbf{E}\|_{1,1} + \|\mathbf{N}\|_F^2\right) \qquad (10)\\ \min_{\mathbf{Z},\mathbf{D},\mathbf{E},\mathbf{N}}\ & \|\mathbf{Z}\|_* + \|\mathbf{S}\|_{2,1} + \alpha\left(\left\|\mathbf{S}^{\mathbf{T}}\mathbf{D}\right\|_F^2 + \|\mathbf{BD}\|_F^2\right) \\ & + \lambda\left(\|\mathbf{E}\|_{1,1} + \|\mathbf{N}\|_F^2\right). \qquad (11)\end{aligned}$$

In the S-model, since the obtained anomaly component $\mathbf{S}$ is sparse distribution, it may contain some sparse outlier information that may be noise or background. To remove the influence of the outlier information, we further optimize the sparse anomaly matrix $\mathbf{S}$ from the S-model, and the optimization manner is from [39] as follows:

$$A^*_{(x_i)} = \left\|\mathbf{S}_{(:,i)} - \mathbf{u_s}\right\|_2^2 \qquad (12)$$

where $\mathbf{S}_{(:,i)}$ denotes the processing of each column of the anomaly part $\mathbf{S}$. $\mathbf{u_s}$ is the mean sparse features for all columns in $\mathbf{S}$. (12) is equivalent to a postprocessing of the final result $\mathbf{S}$. The detection strategy is the same as that described above in (4).

Fig. 2 provides a geometrical interpretation for the hierarchical mutual-incoherence learning strategy. Without losing generality, we consider two 2-D vectors with unit length, i.e., background vector $\mathbf{d}$ and anomaly vector $\mathbf{s}$ with $|\mathbf{d}| = |\mathbf{s}| = 1$, and their inner product is calculated as $\mathbf{d} \bullet \mathbf{s} = \mathbf{d}\mathbf{s}^T = |\mathbf{d}||\mathbf{s}|\cos\theta = \cos\theta$, where θ is the angle between the background and the anomaly vector.

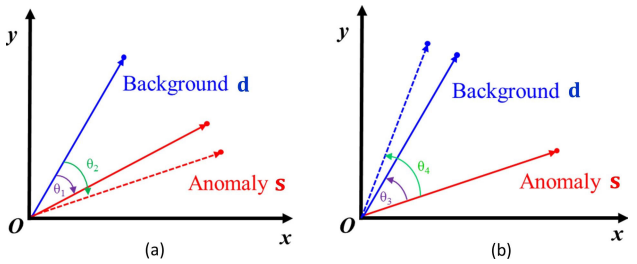


Fig. 2. Geometrical interpretation for the hierarchical mutual-incoherence learning strategy. (a) Optimizing anomaly $\mathbf{s}$ with background $\mathbf{d}$ fixed. (b) Optimizing the background $\mathbf{d}$ with anomaly $\mathbf{s}$ froze.

As shown in Fig. 2, if the background vector $\mathbf{d}$ is frozen, optimizing their incoherence, i.e., $\mathbf{d}^T\mathbf{s}$, can lead to an enlarged angle between them, i.e., $\theta_2 > \theta_1$. Therefore, the discrimination between $\mathbf{d}$ and $\mathbf{s}$ will be enhanced. Similarly, when the anomaly vector $\mathbf{s}$ is frozen, further optimizing the incoherence $\mathbf{s}^T\mathbf{d}$, their discrimination can be further boosted, i.e., $\theta_4 > \theta_3$. As a result, we propose to hierarchically perform the mutual-incoherence optimization, such that the discrimination between the background and anomaly can be gradually strengthened to improve the anomaly distinguishability.

F. Model Optimization

In each alternative updating layer, we need to solve the optimization of the S-model and the D-model. Since $\|\mathbf{D}^T\mathbf{S}\|_F^2 \le \|\mathbf{S}\|_F^2\|\mathbf{D}\|_F^2$, the S-model can be transformed as

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{S},\mathbf{E},\mathbf{N}}\ & \|\mathbf{Z}\|_* + \|\mathbf{S}\|_{2,1} + \alpha'\|\mathbf{S}\|_F^2 + \alpha\|\mathbf{SA}\|_F^2 \\ & + \lambda\left(\|\mathbf{E}\|_{1,1} + \|\mathbf{N}\|_F^2\right) \\ \text{s.t. }\ & \mathbf{X} = \mathbf{DZ} + \mathbf{S} + \mathbf{E} + \mathbf{N} \qquad (13)\end{aligned}$$

where $\alpha' = \alpha\|\mathbf{D}\|_F^2$ is constant.

Similarly, since $\|\mathbf{S}^T\mathbf{D}\|_F^2 \le \|\mathbf{S}\|_\mathbf{F}^2\|\mathbf{D}\|_\mathbf{F}^2$, the D-model can be transformed as

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{D},\mathbf{E},\mathbf{N}}\ & \|\mathbf{Z}\|_* + \|\mathbf{S}\|_{2,1} + \theta'\|\mathbf{D}\|_F^2 + \alpha\|\mathbf{BD}\|_F^2 \\ & + \lambda\left(\|\mathbf{E}\|_{1,1} + \|\mathbf{N}\|_F^2\right) \\ \text{s.t. }\ & \mathbf{X} = \mathbf{DZ} + \mathbf{S} + \mathbf{E} + \mathbf{N} \qquad (14)\end{aligned}$$

where $\theta' = \alpha\|\mathbf{S}\|_F^2$ is constant.

To optimize the low-rank decomposition problem in (13) and (14), two auxiliary variables $\mathbf{J}$ and $\mathbf{L}$ are introduced to separate the variables, so they can be transformed as follows:

$$\begin{aligned}\min_{\mathbf{Z},\mathbf{S},\mathbf{E},\mathbf{N}}\ & \|\mathbf{J}\|_* + \|\mathbf{L}\|_{2,1} + \alpha'\|\mathbf{S}\|_F^2 + \alpha\|\mathbf{SA}\|_F^2 \\ & + \lambda\left(\|\mathbf{E}\|_{1,1} + \|\mathbf{N}\|_F^2\right) \\ \text{s.t. }\ & \mathbf{X} = \mathbf{DZ} + \mathbf{S} + \mathbf{E} + \mathbf{N},\quad \mathbf{J} = \mathbf{Z},\ \mathbf{L} = \mathbf{S} \qquad (15)\\ \min_{\mathbf{Z},\mathbf{D},\mathbf{E},\mathbf{N}}\ & \|\mathbf{J}\|_* + \|\mathbf{L}\|_{2,1} + \theta'\|\mathbf{D}\|_F^2 + \alpha\|\mathbf{BD}\|_F^2 \\ & + \lambda\left(\|\mathbf{E}\|_{1,1} + \|\mathbf{N}\|_F^2\right) \\ \text{s.t. }\ & \mathbf{X} = \mathbf{DZ} + \mathbf{S} + \mathbf{E} + \mathbf{N},\quad \mathbf{J} = \mathbf{Z},\ \mathbf{L} = \mathbf{S}. \qquad (16)\end{aligned}$$

Therefore, (15) and (16) can be solved by ADMM [40], and the augmented Lagrangian functions are expressed as

$$\mathcal{L}_{S\text{-Model}}(\mathbf{Z}, \mathbf{S}, \mathbf{E}, \mathbf{N}, \mathbf{J}, \mathbf{L}, \mathbf{Y}_1, \mathbf{Y}_2, \mathbf{Y}_3, \mu)$$

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$$
\begin{aligned}
&= \|\mathbf{J}\|_* + \|\mathbf{L}\|_{2,1} + \alpha'\|\mathbf{S}\|_F^2 + \alpha\|\mathbf{SA}\|_F^2 + \lambda\|\mathbf{E}\|_{1,1} \\
&\quad + \lambda\|\mathbf{N}\|_F^2 + \langle \mathbf{Y}_1, \mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\rangle + \langle \mathbf{Y}_2, \mathbf{J} - \mathbf{Z}\rangle \\
&\quad + \langle \mathbf{Y}_3, \mathbf{L} - \mathbf{S}\rangle + \frac{\mu}{2}\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_F^2 \\
&\quad + \|\mathbf{J} - \mathbf{Z}\|_F^2 + \|\mathbf{L} - \mathbf{S}\|_F^2 \qquad (17)
\end{aligned}
$$

$$
\begin{aligned}
&\mathcal{L}_{D\text{-Model}}(\mathbf{Z}, \mathbf{D}, \mathbf{E}, \mathbf{N}, \mathbf{J}, \mathbf{L}, \mathbf{Y}_1, \mathbf{Y}_2, \mathbf{Y}_3, \mu) \\
&= \|\mathbf{J}\|_* + \|\mathbf{L}\|_{2,1} + \theta'\|\mathbf{D}\|_F^2 + \alpha\|\mathbf{BD}\|_F^2 + \lambda\|\mathbf{E}\|_{1,1} \\
&\quad + \lambda\|\mathbf{N}\|_F^2 + \langle \mathbf{Y}_1, \mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\rangle + \langle \mathbf{Y}_2, \mathbf{J} - \mathbf{Z}\rangle \\
&\quad + \langle \mathbf{Y}_3, \mathbf{L} - \mathbf{S}\rangle + \frac{\mu}{2}\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_F^2 \\
&\quad + \|\mathbf{J} - \mathbf{Z}\|_F^2 + \|\mathbf{L} - \mathbf{S}\|_F^2 \qquad (18)
\end{aligned}
$$

where the matrix inner product $\langle \mathbf{A}, \mathbf{B}\rangle = \text{trace}(\mathbf{A}^T\mathbf{B})$, $\|\cdot\|_F$ denotes the Frobenious norm, $\mathbf{Y}_1$, $\mathbf{Y}_2$ and $\mathbf{Y}_3$ are Lagrange multipliers, and $\mu > 0$ is a penalty parameter.

The updating process for the S-model is as follows.

1) Updating $\mathbf{J}$ by fixing the other variables and solving the following problem:

$$\mathbf{J} = \underset{\mathbf{J}}{\text{argmin}} \ \|\mathbf{J}\|_* + \langle \mathbf{Y}_2, \mathbf{J} - \mathbf{Z}\rangle + \frac{\mu}{2}\|\mathbf{J} - \mathbf{Z}\|_F^2. \qquad (19)$$

2) Updating $\mathbf{S}$ by fixing the other variables and solving the following problem:

$$
\begin{aligned}
\mathbf{S} = \underset{\mathbf{S}}{\text{argmin}} \ & \beta'\|\mathbf{S}\|_F^2 + \alpha\|\mathbf{SA}\|_F^2 \\
& + \langle \mathbf{Y}_1, \mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\rangle + \langle \mathbf{Y}_3, \mathbf{L} - \mathbf{S}\rangle \\
& + \frac{\mu}{2}\left(\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_F^2 + \|\mathbf{L} - \mathbf{S}\|_F^2\right). \qquad (20)
\end{aligned}
$$

3) Updating $\mathbf{L}$ by fixing the other variables and solving the following problem:

$$\mathbf{L} = \underset{\mathbf{L}}{\text{argmin}} \ \alpha\|\mathbf{L}\|_{2,1} + \langle \mathbf{Y}_3, \mathbf{L} - \mathbf{S}\rangle + \frac{\mu}{2}\|\mathbf{L} - \mathbf{S}\|_F^2. \qquad (21)$$

4) Updating $\mathbf{Z}$ by fixing the other variables and solving the following problem:

$$
\begin{aligned}
\mathbf{Z} = \underset{\mathbf{Z}}{\text{argmin}} \ & \|\mathbf{Z}\|_* \\
& + \langle \mathbf{Y}_1, \mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\rangle + \langle \mathbf{Y}_2, \mathbf{J} - \mathbf{Z}\rangle \\
& + \frac{\mu}{2}\left(\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_F^2 + \|\mathbf{J} - \mathbf{Z}\|_F^2\right). \qquad (22)
\end{aligned}
$$

5) Updating $\mathbf{E}$ by fixing the other variables and solving the following problem:

$$
\begin{aligned}
\mathbf{E} = \underset{\mathbf{E}}{\text{argmin}} \ & \lambda\|\mathbf{E}\|_{1,1} + \langle \mathbf{Y}_1, \mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\rangle \\
& + \frac{\mu}{2}\left(\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_F^2\right). \qquad (23)
\end{aligned}
$$

6) Updating $\mathbf{N}$ by fixing the other variables and solving the following problem:

$$
\begin{aligned}
\mathbf{N} = \underset{\mathbf{N}}{\text{argmin}} \ & \lambda\|\mathbf{N}\|_F^2 + \langle \mathbf{Y}_1, \mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\rangle \\
& + \frac{\mu}{2}\left(\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_F^2\right). \qquad (24)
\end{aligned}
$$

The updating scheme for the D-model is as follows.

Updating $\mathbf{D}$ by fixing the other variables and solving the following problem:

$$
\begin{aligned}
\mathbf{D} = \underset{\mathbf{D}}{\text{argmin}} \ & \theta'\|\mathbf{D}\|_F^2 + \alpha\|\mathbf{BD}\|_F^2 \\
& + \langle \mathbf{Y}_1, \mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\rangle \\
& + \frac{\mu}{2}\left(\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_F^2\right). \qquad (25)
\end{aligned}
$$

The other parameters' optimization is the same as the solving of the S-Model. The solutions of the S-model and the D-model are outlined in Algorithms 1 and 2, where Θ, $\mathcal{S}$, and Ω are the singular value threshold, shrinkage, and $\ell_{2,1}$ minimization operators [40], [41], respectively.

Algorithm 1 Solving S-Model

Input: HSI data set $\mathbf{X} \in \Re^{D\times N}$, Background Dictionary $\mathbf{D} \in \Re^{D\times N_b}$

Initialize: $\mathbf{Z} = \mathbf{J} = \mathbf{S} = \mathbf{L} = \mathbf{E} = \mathbf{N} = \mathbf{Y}_1 = \mathbf{Y}_2 = \mathbf{Y}_3 = 0$, $\mu_{max} = 10^6$, $\mu = 10^{-2}$, $\rho = 1.1$, $\varepsilon = 10^{-6}$

While not converged do

1: Fix others and update $\mathbf{J}$ by

$$\mathbf{J} = \Theta_{1/\mu}(\mathbf{Z} + \mathbf{Y}_2/\mu) \qquad (26)$$

2: Fix others and update $\mathbf{S}$ as (20)

3: Fix others and update $\mathbf{L}$ by

$$\mathbf{L} = \Omega_{\alpha/\mu}(\mathbf{S} + \mathbf{Y}_3/\mu) \qquad (27)$$

4: Fix others and update $\mathbf{Z}$ by

$$\mathbf{Z} = \left(\mathbf{D^T D} + \mathbf{I}\right)^{-1}\begin{pmatrix} \mathbf{J} + \mathbf{D^T}(\mathbf{X} - \mathbf{S} - \mathbf{E} - \mathbf{N}) \\ +\left(\mathbf{D^T Y}_1 - \mathbf{Y}_2\right)/\mu \end{pmatrix} \qquad (28)$$

5: Fix others and update $\mathbf{E}$ by

$$\mathbf{E} = \mathcal{S}_{\lambda/\mu}(\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{N} + \mathbf{Y}_1/\mu) \qquad (29)$$

6: Fix others and update $\mathbf{N}$ as (24)

7: Fix others and update $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$ by

$$
\begin{aligned}
\mathbf{Y}_1{}^* &= \mathbf{Y}_1 + \mu(\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}) \\
\mathbf{Y}_2{}^* &= \mathbf{Y}_2 + \mu(\mathbf{Z} - \mathbf{J}) \\
\mathbf{Y}_3{}^* &= \mathbf{Y}_3 + \mu(\mathbf{S} - \mathbf{L}) \qquad (30)
\end{aligned}
$$

8: Update μ by

$$\mu^* = min(\rho\mu, \mu_{max}) \qquad (31)$$

9: Check the convergence conditions

$$
\begin{aligned}
\|\mathbf{X} - \mathbf{DZ} - \mathbf{S} - \mathbf{E} - \mathbf{N}\|_\infty &< \varepsilon \\
\|\mathbf{Z} - \mathbf{J}\|_\infty &< \varepsilon \\
\|\mathbf{S} - \mathbf{L}\|_\infty &< \varepsilon \qquad (32)
\end{aligned}
$$

End While

Output: $\mathbf{Z}$, $\mathbf{S}$, $\mathbf{E}$, $\mathbf{N}$.

The main computational cost of our algorithm is in matrix inversion and singular value decomposition (SVD). In Algorithm 1, we update $\mathbf{J}$ using SVD, and the complexity of solving $\mathbf{J}$ is about $\mathcal{O}({N_b}^2 N)$. The complexity of solving $\mathbf{L}$ with the $\ell_{2,1}$ minimization operator is about $\mathcal{O}(D^2 N)$.

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Algorithm 2 Solving D-Model

Input: HSI data set $\mathbf{X} \in \Re^{D\times N}$, Anomaly $\mathbf{S} \in \Re^{D\times N}$
$\mathbf{Z} = \mathbf{J} = \mathbf{D} = \mathbf{L} = \mathbf{E} = \mathbf{N} = \mathbf{Y}_1 = \mathbf{Y}_2 =$
Initialize: $\mathbf{Y}_3 = 0$, $\mu_{max} = 10^6$, $\mu = 10^{-2}$, $\rho = 1.1$,
$\varepsilon = 10^{-4}$, $\eta = \alpha\|\mathbf{B}\|_F^2$
While not converged do

1: Fix others and update $\mathbf{J}$ as (26)
2: Fix others and update $\mathbf{L}$ as (27)
3: Fix others and update $\mathbf{D}$ by

$$\mathbf{D} = \left((\mathbf{X} - \mathbf{S} - \mathbf{E} - \mathbf{N}) + \frac{\mathbf{Y}_1}{\mu}\right)\mathbf{Z}^{\mathbf{T}} \times \left(\mathbf{Z}\mathbf{Z}^{\mathbf{T}} + \left(\frac{2\theta' + \eta}{\mu}\right)\mathbf{I}\right)^{-1} \quad (33)$$

4: Fix others and update $\mathbf{Z}$ as (28)
5: Fix others and update $\mathbf{E}$ by (29)
6: Fix others and update $\mathbf{N}$ as (24)
7: Fix others and update $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$ as (30)
8: Update μ as (31)
9: Check the convergence conditions as (32)
End While
Output: $\mathbf{Z}$, $\mathbf{D}$, $\mathbf{E}$, $\mathbf{N}$.

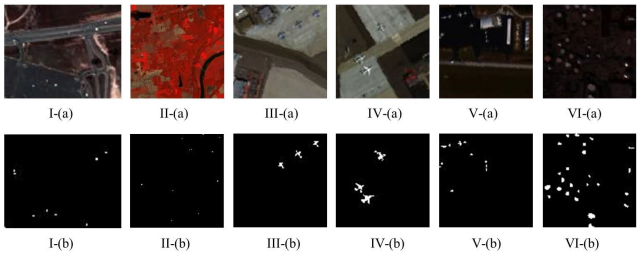


Fig. 3. Experimental datasets. (a) Pseudocolor image. (b) Ground truth. (I) HYD. (II) Hyp. (III) SD1. (IV) SD2. (V) Ga. (VI) LA.

The main computational cost of solving $\mathbf{Z}$ is the matrix inverse computation with $\mathcal{O}({N_b}^3)$. The computational complexity of solving $\mathbf{E}$ by the shrinkage operator is about $\mathcal{O}({N_b}^2DN)$. Therefore, the total computational complexity of Algorithm 1 is about $\mathcal{O}(\omega({N_b}^2N + D^2N + {N_b}^3 + {N_b}^2DN))$. Similarly, the total computational complexity of Algorithm 2 is approximately $\mathcal{O}(\omega({N_b}^2N + D^2N + 2{N_b}^3 + {N_b}^2DN))$, where ω is the number of iterations.

IV. Experimental Results and Analysis

A. Dataset Description

In the experiments, we validated the performance of the AHMID model. All the experiments were implemented using MATLAB 2020a on a computer with i5-10400F CPU. The datasets are shown in Fig. 3 including their pseudocolor images and ground truth.

1) *HYDICE Urban (HYD) Dataset [19]:* The first dataset was captured by the Hyperspectral Digital Image Acquisition Experiment (HYDICE) on an aircraft platform with a spectral resolution of 10 nm and a spatial resolution of about 1 m. The spatial size used in the experiment is 80 × 100 and has 175 spectral bands, with 162 bands remaining after removing low-SNR and vapor absorption bands. There are 19 anomalous pixels in the HYD dataset for detection.
2) *Hyperion (Hyp) Dataset [32]:* The second real data were acquired by the Hyp imaging sensor, covering an agricultural area of the State of Indiana, USA, in 2008. This dataset has 242 bands with a spectral resolution of 10 nm, and 149 bands are left after removing the low-SNR bands and the uncalibrated bands. The 150 × 150 image has 17 anomalies, where the anomalies come from "storage silo" and "roof."
3) *San Diego (SD) Dataset [30]:* The third and fourth HSIs were captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in San Diego, CA, USA with a spatial resolution of 3.5 m and a wavelength range of 370–2510 nm. The spatial size used in the experiment is 100 × 100 and has 189 spectral bands, which has removed the low-SNR, absorbing regions, and bad bands. These two HSIs are called SD-1 and SD-2. The airports with 58 pixels in SD1 and 134 pixels in SD2 are considered as anomalies.
4) *Gainesville (Ga) Dataset [42]:* The fifth HSI was also acquired by the AVIRIS sensor, which has a spatial resolution of 3.5 m. This image has 100 × 100 pixels covering the Gainesville area, FL, USA. One hundred and ninety one bands remained after removing water vapor absorption bands. There are 52 anomalous pixels.
5) *Los Angeles (LA) Dataset [42]:* The sixth HSI was also acquired by the AVIRIS sensor covering the Los Angeles city area with 100 × 100 pixels and 205 spectral bands. The spatial resolution is 7.1 m. The buildings with 272 pixels belong to anomaly targets.

B. Experimental Setup

In this article, a total of eight representative target detection methods are used for comparison, including the statistics-based methods such as the global RX [12], [13] and the segmented RX, the representation-based methods such as the CRD [19], the LRR-based methods such as LRASR [30], MLW_LRRSTO [31], and GTVLRR [32], and the LSDM-based methods such as RPCA-RX [43], LSMAD [28], and LSDM-MoG [24].

In the proposed AHMID, the dictionary $\mathbf{D}$ plays an important role in representing the background pixels. In this article, we adopt the dictionary construction strategy (DCS) from [30], where it first divides the pixels into K clusters using K-means, then calculates the Mahalanobis distance between each test pixel and the local background mean in each cluster, and finally uses the pixels with the smallest distance as the atoms of the dictionary $\mathbf{D}$.

The parameters of the compared algorithm on different datasets are shown in Table I. For CRD, it has the dual windows $(\omega_{\text{in}}, \omega_{\text{out}})$ and the tradeoff parameter λ. For LRASR [30], it includes the number of clusters K, the number of selected atoms per cluster P, and the weighting parameters on ℓ_1-norm as β and $\ell_{2,1}$-norm as λ. MLW_LRRSTO has two tradeoff parameters η and λ. GTVLRR contains three tradeoff

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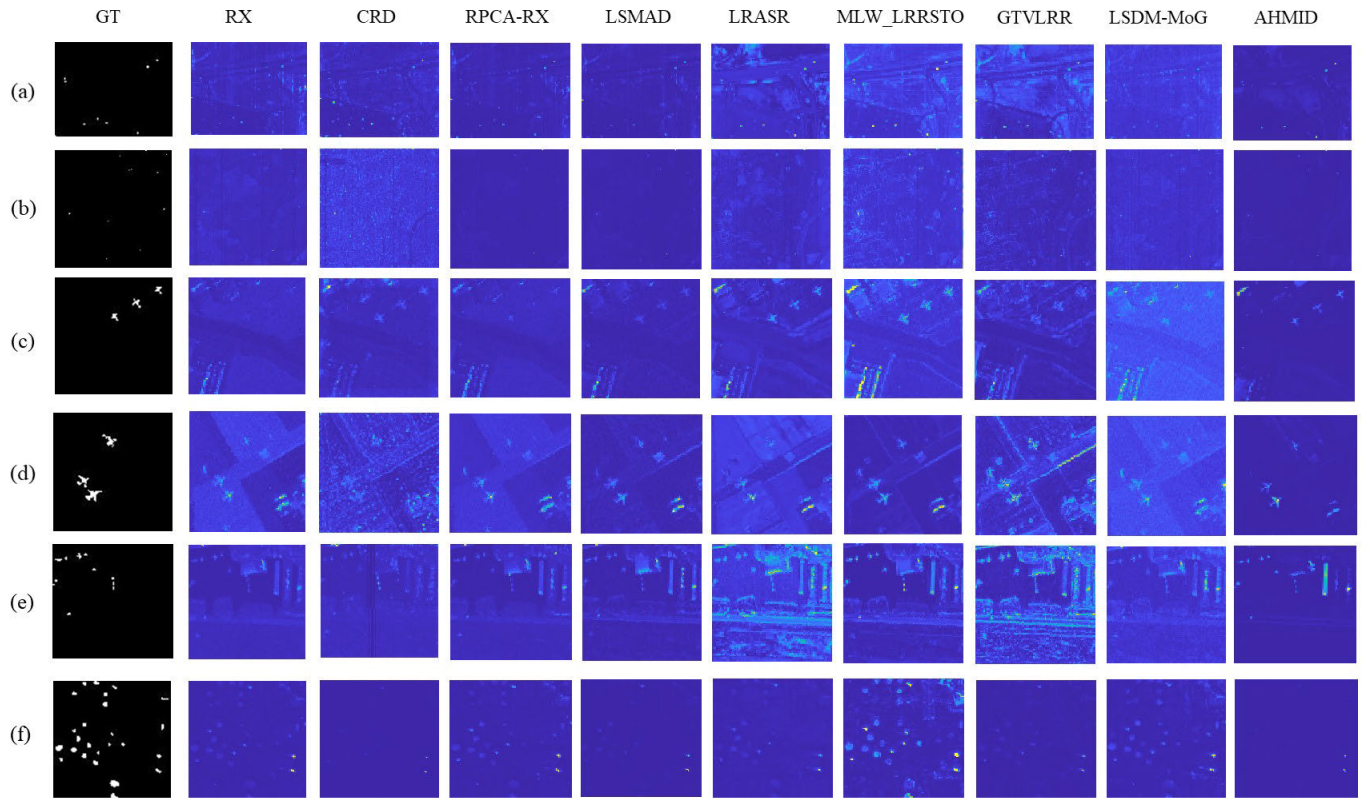


Fig. 4. Detection maps of different methods. (a) HYD. (b) Hyp. (c) SD1. (d) SD2. (e) Ga. (f) LA.

TABLE I
PARAMETERS OF ALL THE ALGORITHMS

Dataset		HYD	Hyp	SD1	SD2	Ga	LA
CRD	ω_{in}	3	3	15	3	3	3
	ω_{out}	5	5	17	5	5	5
	λ	0.01	0.01	0.01	0.01	0.01	0.01
LSMAD	r	28	28	28	28	28	28
	k	0.002	0.002	0.002	0.002	0.002	0.002
LRASR	K	15	15	15	15	15	15
	P	20	20	20	20	20	20
	λ	0.1	1	0.1	0.1	1	0.1
	β	0.1	0.1	0.1	0.1	0.1	0.1
MLW_LRRSTO	η	1	1	1	1	1	1
	λ	0.1	0.1	0.1	0.1	0.1	0.1
GTVLRR	λ	0.5	0.5	0.5	0.5	0.5	0.5
	β	0.2	0.2	0.2	0.2	0.2	0.2
	γ	0.05	0.05	0.05	0.05	0.05	0.05
LSDM-MoG	l_0	2	2	2	2	2	2
	K	10	10	10	10	10	10
AHMID	α	1	1	1	1	1	1
	λ	0.1	0.05	0.01	0.01	0.1	0.05

parameters λ, β, and γ. LSDM-MoG has two parameters including the initial rank l_0 and the number of mixture Gaussian noise K.

To measure the detection performance of different AD methods, the receiver operating characteristic (ROC) is widely used for intuitive observation. We use the ROC curves of (P_f, P_d) and (τ, P_f) to represent the overall detection effectiveness and the background suppression ability, respectively. P_f and P_d are related to τ, and they can be obtained by

$$P_d^{(\tau)} = \frac{\mathrm{TP}^{(\tau)}}{N_{\mathrm{tgt}}}, \quad P_f^{(\tau)} = \frac{\mathrm{FP}^{(\tau)}}{N_{\mathrm{bkg}}} \tag{34}$$

where $\mathrm{TP}^{(\tau)}$ and $\mathrm{FP}^{(\tau)}$ denote the number of target and background pixels whose detection values are greater than a given threshold τ, and N_{tgt} and N_{bkg} are the total number of target and background pixels.

C. Experimental Results and Analysis

Fig. 4 depicts the detection results of different AD algorithms on the six HSI datasets. From Fig. 4, CRD and RPCA-RX can only detect a small number of anomaly targets. MLW_LRRSTO and LSDM-MoG can detect most of the anomaly targets but with a high false alarm rate. For example, in the third and fourth rows of the figure, most of the comparison methods treat many background components as the anomaly targets. For the proposed AHMID, it suppresses the background while maintaining the detection of more anomaly pixels. For the Ga dataset, most methods fail to detect the target, especially LRASR and GTVLRR, and although they detect most of the targets, a lot of background pixels are misclassified as target pixels. Observing the aircraft in the third row, the compared AD algorithms do not either detect the anomaly targets or miss the detected target shapes, but the proposed AHMID algorithm can fully maintain the detected target shapes.

To comprehensively compare the detection accuracy of the methods, the ROC curves of (P_f, P_d) and (τ, P_f) for each dataset are given in Figs. 5 and 6. From the ROC curves of (P_f, P_d), AHMID has a greater detection probability than the other methods on the six datasets under different false alarm rates. For the Ga dataset, MLW_LRRSTO has the largest

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TABLE II
AUC VALUES OF THE METHODS FOR THE SIX DATASETS

$AUC_{(P_f,\ P_d)}$	HYD	Hyp	SD1	SD2	Ga	LA	Average
RX	0.9824	0.9978	0.8885	0.9519	0.9513	0.9887	0.9601
CRD	0.9892	0.9933	0.9798	0.9236	0.9359	0.8973	0.9532
RPCA-RX	0.9803	0.9970	0.9165	0.9437	0.9577	0.9871	0.9637
LSMAD	0.9925	0.9984	0.9752	0.9774	0.9626	0.9814	0.9813
LRASR	0.9944	0.9997	0.9899	0.9168	0.9011	0.9652	0.9612
MLW_LRRSTO	0.9961	0.9963	0.9810	0.9830	0.9790	0.9765	0.9853
GTVLRR	0.9939	0.9996	0.9632	0.9556	0.9221	0.9557	0.9650
LSDM-MoG	0.9657	0.9959	0.9067	0.9431	0.9292	0.9786	0.9532
AHMID	**0.9985**	**0.9998**	**0.9923**	**0.9891**	**0.9845**	**0.9964**	**0.9934**
$AUC_{(\tau,\ P_f)}$	HYD	Hyp	SD1	SD2	Ga	LA	Average
RX	0.0345	0.0331	0.0381	0.0707	0.0351	0.0114	0.0372
CRD	0.0339	0.0824	0.0372	0.0709	0.0372	0.0015	0.0439
RPCA-RX	0.0210	0.0113	0.0347	0.0604	0.0327	0.0115	0.0286
LSMAD	0.0183	0.0105	0.0239	0.0272	0.0315	0.0017	0.0189
LRASR	0.0606	0.0443	0.0450	0.0830	0.1444	0.0271	0.0674
MLW_LRRSTO	0.0298	0.0194	0.0191	0.0204	0.0337	0.0038	0.0210
GTVLRR	0.0922	0.0336	0.0398	0.0757	0.1214	0.0094	0.0620
LSDM-MoG	0.0782	0.0528	0.1626	0.1237	0.0742	0.0186	0.0850
AHMID	**0.0105**	**0.0056**	**0.0080**	**0.0060**	**0.0107**	**0.0001**	**0.0068**

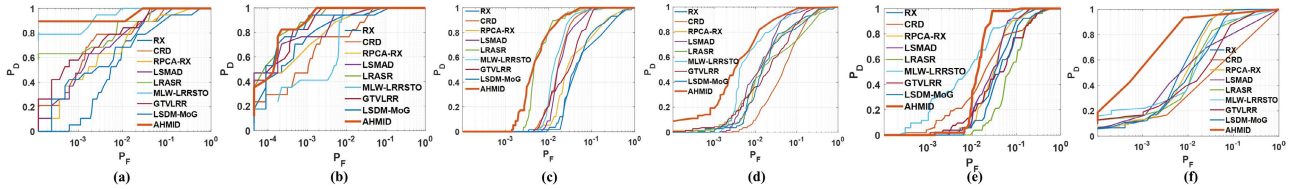


Fig. 5. ROC of (P_f, P_d) for different methods. (a) HYD. (b) Hyp. (c) SD1. (d) SD2. (e) Ga. (f) LA.

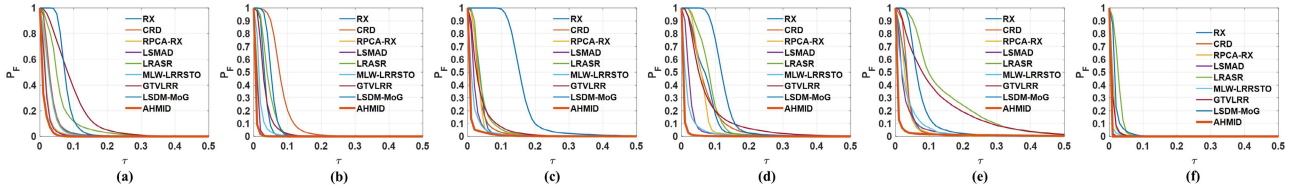


Fig. 6. ROC of (τ, P_f) for different methods. (a) HYD. (b) Hyp. (c) SD1. (d) SD2. (e) Ga. (f) LA.

detection probability when the false alarm rate is less than 0.01, whereas AHMID has the largest detection probability when the false alarm rate is greater than 0.01. From the ROC curves of (τ, P_f), AHMID has a lower detection probability under different thresholds τ, which indicates that the proposed method generates stronger suppression performance for background on all the six datasets to show the consistent visual detection results, as shown in Fig. 4.

Table II shows the AUC values from the ROC curves. $\text{AUC}_{(P_f,P_d)}$ reflects the overall detection accuracy, whereas $\text{AUC}_{(\tau,P_f)}$ responds to the background suppression accuracy. From Table II, AHMID achieves the best $\text{AUC}_{(P_f,P_d)}$ values and the smallest $\text{AUC}_{(\tau,P_f)}$ values compared with the other methods on the six datasets. The results indicate that the proposed method has excellent AD and background suppression performance. The main reason is that the proposed method considers the discrimination of features between the background and the anomalies, the modeling of the mixed noise, and the hierarchical alternative updating to enhance the representation of background and anomaly targets.

D. Discussion and Analysis

1) Separability Analysis: To analyze the separability of background and anomalies, Fig. 7 shows the separation box-plots between the anomalies and the background according to the detection values, which reveals the separability of the anomaly targets from the background pixels to measure the detection performance. In the figure, the red box-plots mean the anomaly target, and the blue box-plots indicate the background. In general, these methods have different separation effects on different datasets. If the margin between the background and anomalies is large, the anomaly pixels can be easily detected from the background, indicating that the detector has good performance. The box-pot between the anomalies and the background has a lot of overlap, such as LRASR and GTVLRR on the Ga dataset, which indicates that the anomaly separation ability is limited. For background, a small box-pot explains that the background is relatively concentrated, which indicates that the background suppression is very strong. For AHMID, there is almost no overlap between the background and anomalies on the six datasets, and the background is very compact. The results show that

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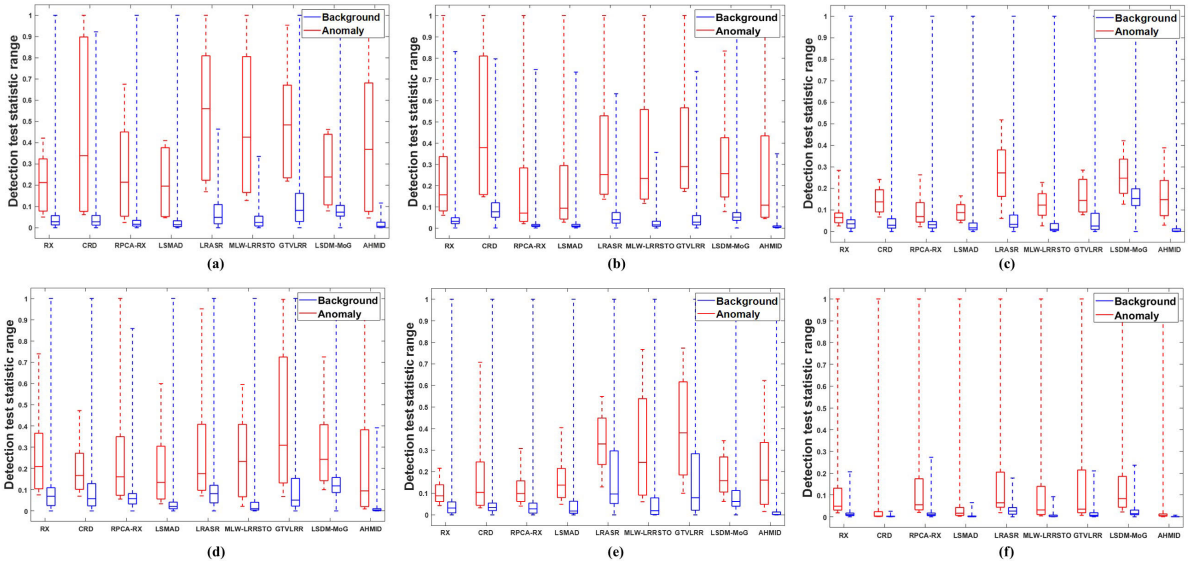


Fig. 7. Separation box-plots of different methods. (a) HYD. (b) Hyp. (c) SD1. (d) SD2. (e) Ga. (f) LA.

TABLE III
ABLATION EXPERIMENTS ON THE HYD DATASET

Model	TFS	BDS	MNM	$AUC_{(P_f,\ P_d)}$	$AUC_{(\tau,\ P_f)}$
M1	✓	✓	✓	**0.9985**	**0.0105**
M2		✓	✓	0.8515	0.0504
M3	✓		✓	0.9855	0.0282
M4	✓	✓		0.9904	0.0157

the proposed method processes strong background suppression and anomaly separation ability.

2) Ablation Studies: The proposed AHMID method is based on the structural incoherence to add the target first-order statistic (TFS), the background decentralized suppression (BDS), and the mixed noise modeling (MNM). We conducted the ablation analysis on the HYD dataset to validate the effectiveness of each part. Table III shows the results of the ablation experiments. M1 denotes the completed model of AHMID, and the ablation methods achieve weaker results than the completed AHMID in terms of $\text{AUC}_{(P_f,\ P_d)}$ and $\text{AUC}_{(\tau,\ P_f)}$. Comparing M1 and M2, the removal of the TFS module has a substantially negative effect on the detection effect, which can enhance the discrimination of anomaly features with the first-order statistic on the sparse anomalies. Similarly, for M1 and M3, the decentralization of the background dictionary can improve the performance of the background suppression. In addition, the MNM is beneficial to remove the influence of complex noise from the imaging environment according to M1 and M4. These results indicate that the proposed models are both effective to enhance the targets and suppress the background.

3) Hierarchical Analysis: The effectiveness of the hierarchical strategy of the proposed method is explored on the HYD dataset shown in Fig. 8. From the detection map in Fig. 8(a), it can be visually observed that the results of the first layer without the hierarchical strategy (layer = 0) are much worse compared with the first layer with the hierarchical strategy.

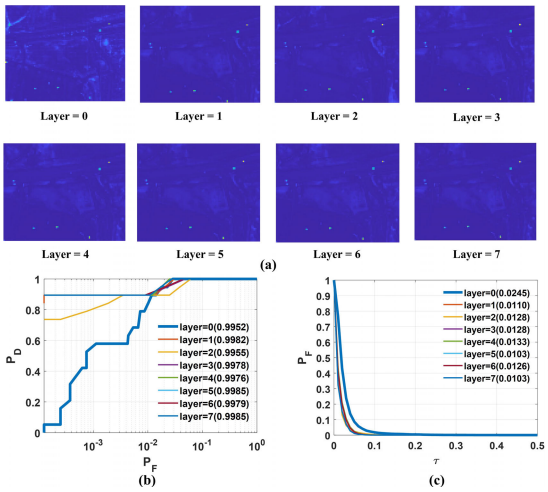


Fig. 8. Results under different layers on the HYD dataset. (a) Detection maps. (b) ROC of (P_f, P_d). (c) ROC of (τ, P_f).

The results of the first layer are clearly stronger than layer = 0 in terms of background suppression and more highlighted targets. A better detection performance can be achieved with five layers. The anomaly targets are better detection and the background is better suppression compared with the other layers. From Fig. 8(b) and (c) curves, it can be found that too many layers may lead to a bias in the separation of the background and anomalies. For example, the second layer has the worst overall detection result compared with the other layers, and although the fifth and seventh layers have the best results, they will consume more time. Therefore, the influence of the number of layers on the detection results is also crucial, and the number of layers tends to achieve the best detection within an appropriate number of layers.

4) Mixed Noise Analysis: To validate the effectiveness of our method for modeling mixed noise with $\ell_{1,1}$-norm and Frobenius norm, we added Gaussian noise (G) and strip

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TABLE IV
AUC VALUES WITH MIXED NOISE ON THE HYD DATASET

$AUC_{(P_f,\ P_d)}$	No noise	G	S	G+S
RX	0.9824	0.9637	0.9535	0.8368
CRD	0.9892	0.9714	0.9687	0.9248
RPCA-RX	0.9803	0.9394	0.9277	0.8206
LSMAD	0.9925	0.9955	0.8675	0.8342
LRASR	0.9944	0.9647	0.7125	0.7643
MLW_LRRSTO	0.9961	0.9918	0.7302	0.7181
GTVLRR	0.9939	0.9905	0.9079	0.8194
LSDM-MoG	0.9657	0.8881	0.4559	0.4793
AHMID	**0.9985**	**0.9959**	**0.9944**	**0.9457**
$AUC_{(\tau,\ P_f)}$	No noise	G	S	G+S
RX	0.0345	0.0870	0.1482	0.2880
CRD	0.0339	0.0345	0.0587	0.1764
RPCA-RX	0.0210	0.0386	0.0608	0.2847
LSMAD	0.0183	0.0548	0.2059	0.2322
LRASR	0.0606	0.1441	0.3100	0.3305
MLW_LRRSTO	0.0298	0.0662	0.3740	0.3506
GTVLRR	0.0922	0.1139	0.2623	0.3614
LSDM-MoG	0.0782	0.3717	0.3614	0.2573
AHMID	**0.0105**	**0.0239**	**0.0537**	**0.0623**

TABLE V
AUC VALUES WITH DIFFERENT DCSS

DCS	$AUC_{(P_f,\ P_d)}$			$AUC_{(\tau,\ P_f)}$		
	LRASR	GTVLRR	AHMID	LRASR	GTVLRR	AHMID
$\mathbf{D}^r$	0.9781	0.9566	0.9928	0.2161	0.1432	0.0116
$\mathbf{D}^c$[30]	0.9847	0.9932	0.9985	0.0696	0.0902	0.0105

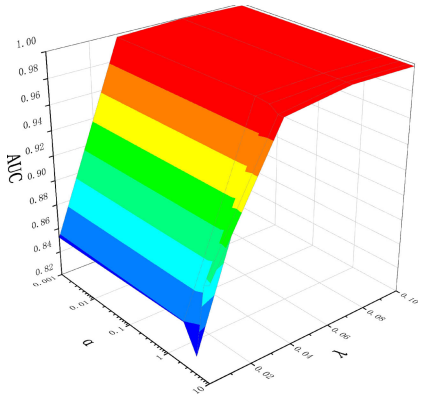


Fig. 9. AUC values of AHMID with different parameters on the HYD dataset.

TABLE VI
RUNNING TIME OF DIFFERENT METHODS (s)

Dataset	HYD	Hyp	SD1	SD2	Ga	LA
RX	0.074	0.113	0.064	0.083	0.065	0.085
CRD	0.972	2.669	7.039	1.585	1.249	1.604
RPCA-RX	2.038	6.464	3.228	3.409	4.042	8.945
LSMAD	6.273	19.188	10.302	10.620	11.616	14.858
LRASR	21.537	87.071	30.417	109.762	40.734	52.679
MLW_LRRSTO	34.940	113.112	50.003	52.980	49.087	42.470
GTVLRR	123.287	323.006	133.079	268.752	174.259	158.675
LSDM-MoG	5.938	14.277	5.780	5.976	13.159	15.401
AHMID	26.189	35.697	45.148	102.788	13.845	47.932

noise (S) with an SNR of 30 dB to the HYD dataset, as shown in Table IV. Compared with the other eight benchmark methods, the previous LRR-based detection methods either do not consider the effect of noise or only consider a single Gaussian noise, which would perform poorly in the face of complex noise situations. Methods such as LRASR and MLW_LRRSTO do not deal with noise. The performance of the LRR-based detector is significantly worse in the face of Gaussian noise. Although MLW_LRRSTO does not consider noise alone, the spatial constraints are added, and the spatial information becomes important with the addition of noise. Therefore, MLW_LRRSTO also performs well with the Gaussian noise. In addition, GTVLRR has a graph regularizer that also has good suppression of the Gaussian noise. However, with the addition of stripe noise, the performance of these three LRR-based detectors decreases significantly, especially when the Gaussian noise and the stripe noise are mixed. The proposed method can consistently achieve advantageous detection performance even though there is serious data pollution with the mixed noise. The benefits can be attributed to the excellent MNM via the noise constraint term with $\ell_{1,1}$-norm and Frobenius norm. As a result, the three basic components are hierarchically and alternately optimized to obtain a global optimal solution, which endows the proposed model with good antinoise ability.

5) Dictionary Construction: The background dictionary plays an important role in the subsequent detection. To validate the effectiveness of different DCSs, we compared two dictionary generation manners including predetection dictionary $\mathbf{D}^c$ from [30] and randomly selected dictionary $\mathbf{D}^r$ from data. We compared the proposed method with two dictionary-based methods (LRASR and GTVLRR). Table V shows the results of 10 repeated experiments. The proposed method is more stable and achieves better detection effect, and the randomly selected dictionary has a big influence on the detection results for LRASR and GTVLRR compared with our method. The results further illustrate that the initial background dictionary selection for AHMID is not very demanding.

6) Parameter Analysis: AHMID includes the two tradeoff parameters α and λ. We evaluate the effects of parameters on the HYD dataset shown in Fig. 9. To ensure the consistency of the experiment, we chose the number of layers as 1. From Fig. 9, the proposed method is insensitive to α with a subtle change on AUC in a large range of α. The AUCs increase when λ goes from 0.01 to 0.1. A better AUC can be obtained when λ exceeds 0.05. Therefore, the detection performance is effective with a reasonable value for the tradeoff parameters of the proposed method.

7) Running Time: We also compared the running time of all the above methods on different datasets, and the detailed results are shown in Table VI. RX is a simple model and performs more efficiently than the other methods. The CRD method uses the double windows to estimate the local background of the measured pixels and is more computational complexity than RX. The decomposition or representation-based methods require more running time than RX. The LRASR and GTVLRR methods spend more time on dictionary construction and iterative solving. Since our method also requires iterative optimization, it has no significant advantage in terms of running time than most methods. However, our method has lower cost than GTVLRR.

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V. Conclusion

In this article, we propose an AD method for HSI called AMHID. Unlike the previous LRR-based methods, AMHID utilizes the structural incoherence constraint to separate background and anomaly. Simultaneously, it uses the first-order statistic and background decentralization to constrain the target and the background, which can further enhance the anomalies and suppress the background. Meanwhile, we construct a mixed noise model with $\ell_{1,1}$-norm and Frobenius norm to remove the influence of noise. To improve the representation of the background and anomalies, a hierarchical alternative optimization idea is designed by constructing an S-model with a fixed background dictionary to update anomalies, as well as a D-model with fixed anomalies to update the background dictionary. In addition, the proposed method does not require a high-quality preconstructed background dictionary, and a poor initial background dictionary can be optimized in the hierarchical alternative updating. Extensive experiments have verified the benefits of the proposed method on six benchmark datasets compared with many state-of-the-art methods.

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